

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

|                |                                                                                                                                                 |               |                                                         |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------|
| Course Code:   | CSA10 / CSA1063                                                                                                                                 | Course Title: | Software Engineering                                    |
| CO Assessed:   | CO2                                                                                                                                             | Assessment:   | Assessment Tool 2 – Scenario-Based Requirement Analysis |
| Weightage:     | 20%                                                                                                                                             | Total Marks:  | 25                                                      |
| CO2 Statement: | <i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i> |               |                                                         |
| Student Name:  | S.PRATEEKSHA                                                                                                                                    | Reg. No.:     | 192572137                                               |
| Date:          | 3\8\26                                                                                                                                          | Duration:     | 60 minutes                                              |

### Code of Conduct

*I S.PRATEEKSHA-192572137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S.PRATEEKSHA

### Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

### Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

#### 1. Study and Analyze the Scenario

- Carefully understand the given problem statement.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

The project belongs to the **Human Resource Management (HRM)** and **Business Analytics** domain. It helps organizations evaluate employee performance, monitor productivity, generate analytical reports, and support fair performance appraisal.

#### 1.2 Stakeholders

- Company Management
- Human Resource (HR) Department
- Team Managers/Supervisors
- Employees
- System Administrator
- IT Support Team

#### 1.3 Existing Challenges

Many organizations still rely on manual spreadsheets or traditional evaluation methods, leading to several issues:

- Time-consuming performance evaluation process.
- Lack of real-time productivity tracking.
- Human bias during employee appraisal.
- Difficulty in generating performance reports.
- Poor data organization and record management.
- Limited employee feedback and transparency.
- Inefficient decision-making regarding promotions and training.

## 1.4 System Objectives

The proposed system aims to:

- Automate employee performance evaluations.
- Track employee productivity continuously.
- Generate performance analytics dashboards.
- Provide accurate appraisal reports.
- Improve transparency in employee evaluation.
- Assist management in making data-driven HR decisions.

## 2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

### 2.1 Functional Requirements

#### User Management

- User Registration
- Secure Login
- Role-Based Access Control
- Password Reset
- User Profile Management

#### Employee Management

- Add Employee
- Update Employee Details
- Delete Employee
- View Employee Information
- Search Employees

#### Performance Evaluation

- Create Evaluation Criteria
- Assign Performance Scores
- Record Manager Feedback
- Employee Self-Assessment
- Final Performance Rating

## Productivity Analytics

- Track Attendance
- Monitor Task Completion
- Measure Productivity Score
- Generate Performance Trends
- Compare Team Performance

## Reports

- Generate Monthly Reports
- Generate Annual Reports
- Export Reports (PDF/Excel)
- Performance Dashboard

## Notification Module

- Evaluation Reminder
- Report Notifications
- Feedback Alerts

## Administration

- Manage Users
- Manage Departments
- Configure Evaluation Parameters
- Backup Database

Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

## 2.2 Non-Functional Requirements

### Performance

- Dashboard should load within **3 seconds**.
- Support **500+ concurrent users**.
- Reports generated within **5 seconds**.

## **Security**

- Secure Login Authentication
  - Password Encryption
  - Role-Based Authorization
  - Secure Database Access
  - Audit Logs

## **Reliability**

- System uptime of **99.9%**
- Automatic Backup
- Error Recovery Mechanism

## **Usability**

- User-friendly Interface
- Easy Navigation
- Responsive Design
- Simple Dashboard

## **Scalability**

- Easily add new employees.
- Support multiple branches.
- Handle increasing organizational data.

## **Maintainability**

- Modular architecture
- Easy software updates
- Well-documented code

## **Availability**

- 24×7 system availability
- Cloud database support
- Disaster recovery mechanism

### 3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

#### **Primary Actors**

- Employee
- HR Manager
- Team Manager
- System Administrator

#### **Secondary Actors**

- Company Management
- Database Server
- Email Notification Service

- List the major use cases associated with each actor.

#### **Major Use Cases**

##### **Employee**

- Login
- View Profile
- View Performance Score
- Complete Self-Assessment
- View Feedback
- Download Reports

##### **HR Manager**

- Manage Employees
- Configure Evaluation Criteria
- Generate Reports
- View Analytics
- Monitor Performance

##### **Team Manager**

- Evaluate Employees
- Assign Ratings
- Submit Feedback

- View Team Productivity

### **Administrator**

- Manage Users
- Manage Roles
- Configure System
- Backup Database
- Monitor System

### **Management**

- View Organizational Reports
- Analyze Productivity
- Review Performance Trends

- Draw a Use Case Diagram representing the interactions between actors and the system.

## **4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

### **4.1 Introduction**

The Employee Performance Evaluations and Productivity Analytics System is designed to automate employee evaluation processes and improve organizational productivity. The system enables managers to assess employee performance, monitor productivity, generate reports, and make informed HR decisions through data analytics.

### **4.2 Problem Description**

Traditional employee evaluation methods are manual, inconsistent, and time-consuming. Organizations face challenges such as inaccurate performance assessment, delayed reporting, lack of transparency, and inefficient productivity monitoring. These issues reduce employee satisfaction and affect organizational growth.

### **4.3 Scope of the System**

The system will:

- Manage employee records.
- Conduct performance evaluations.
- Monitor productivity metrics.

- Generate analytical reports.
- Provide dashboards.
- Support decision-making.
- Improve transparency.
- Reduce manual work.

#### **4.4 Functional Requirements**

The system shall:

- Authenticate users securely.
- Maintain employee records.
- Conduct performance evaluations.
- Record manager feedback.
- Allow self-assessment.
- Calculate performance scores.
- Generate reports.
- Display productivity dashboards.
- Send notifications.
- Maintain audit logs.

#### **4.5 Non-Functional Requirements**

The system shall provide:

- High Performance
- Strong Security
- Reliability
- Easy Usability
- Scalability
- Maintainability
- High Availability

#### **4.6 Use Case Descriptions**

##### **Use Case 1: Login**

**Actor:** Employee, HR Manager, Team Manager, Administrator

**Description:** Users authenticate themselves using their credentials.

**Pre-condition:**

User account exists.

**Post-condition:**

User accesses the dashboard.

**Use Case 2: Evaluate Employee**

**Actor:** Team Manager

**Description:**

Manager evaluates employee performance based on predefined criteria.

**Pre-condition:**

Employee exists.

**Post-condition:**

Performance score is stored.

**Use Case 3: Generate Report**

**Actor:** HR Manager

**Description:**

Generate performance reports for selected employees or departments.

**Pre-condition:**

Evaluation data available.

**Post-condition:**

Report generated successfully.

**Use Case 4: View Dashboard**

**Actor:** Management

**Description:**

Displays analytics, productivity charts, KPIs, and performance trends.

**Pre-condition:**

Performance data available.

**Post-condition:**

Dashboard displayed.

## 4.7 Data Requirements

The system stores the following data:

### Employee Data

- Employee ID
- Name
- Department
- Designation
- Contact Information

### Evaluation Data

- Evaluation ID
- Evaluation Date
- Ratings
- Feedback
- Final Score

### Productivity Data

- Attendance
- Task Completion
- Working Hours
- Productivity Score

### User Data

- Username
- Password (Encrypted)
- Role
- Login History

## **Report Data**

- Monthly Reports
- Annual Reports
- Performance Charts
- Productivity Statistics

## **4.8 External Interface Requirements**

### **User Interface**

- Responsive Web Dashboard
- Mobile-Friendly Interface
- Interactive Charts
- Easy Navigation

### **Hardware Interface**

- Computer
- Laptop
- Smartphone
- Printer

### **Software Interface**

- Web Browser
- Database Server
- Email Server
- Cloud Storage

### **Communication Interface**

- HTTPS
- Email Notifications
- Internet Connection

## **4.9 System Features**

- Employee Registration
- Secure Authentication

- Employee Profile Management
- Performance Evaluation
- Self Assessment
- Productivity Monitoring
- Dashboard Analytics
- Report Generation
- Notification System
- Role-Based Access Control
- Database Backup
- Audit Logging

## 5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

### 5.1 Assumptions

- Employees and managers have valid user accounts.
- Users have internet access.
- Employee information is accurate and regularly updated.
- Managers complete evaluations within scheduled timelines.
- The organization provides necessary hardware and software resources.
- Users have basic computer literacy.
- The database is regularly maintained and backed up.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

### 5.2 Constraints

#### Technical Constraints

- Requires stable internet connectivity.
- Depends on database server availability.
- Browser compatibility must be ensured.

#### Operational Constraints

- Users require initial training.
- Evaluation policies vary across organizations.
- Continuous data updates are required.

### **Economic Constraints**

- Development budget limitations.
- Licensing costs for cloud services.
- Maintenance expenses.

### **Legal Constraints**

- Compliance with employee data privacy regulations.
- Secure handling of confidential employee information.
- Role-based access to sensitive records.

### **Time Constraints**

- Limited project development schedule.
- Timely completion of testing and deployment.
- Regular updates during evaluation cycles.

## **6. Validate Requirement Completeness and Consistency**

- Verify whether all stakeholder requirements have been addressed

### **6.1 Requirement Completeness**

The SRS addresses the needs of all major stakeholders, including employees, HR managers, team managers, administrators, and company management. It covers authentication, employee management, performance evaluation, productivity tracking, reporting, notifications, security, and analytics.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

### **6.2 Consistency Check**

The requirements are consistent, with no conflicting functionalities between user roles. Functional requirements align with the system objectives, and non-functional requirements support performance, reliability, usability, and security goals.

### **6.3 Ambiguity and Redundancy Check**

The requirements use clear terminology, minimizing ambiguity. Duplicate functionalities have been removed, and each requirement has a distinct purpose.

- Suggest improvements to enhance the quality and completeness of the requirement specification.

## 6.4 Suggested Improvements

- Integrate AI-based performance prediction and personalized recommendations.
- Provide customizable KPI and evaluation templates for different departments.
- Add real-time productivity dashboards with advanced visualizations.
- Enable integration with payroll, attendance, and project management systems.
- Develop a mobile application for employee self-service and manager approvals.
- Implement multilingual support to improve accessibility.
- Incorporate employee sentiment analysis using feedback surveys.

## Conclusion

The **Employee Performance Evaluations and Productivity Analytics System** provides a comprehensive solution for automating performance appraisal and productivity monitoring. By replacing manual evaluation processes with a secure, analytics-driven platform, the system improves transparency, reduces administrative effort, enhances decision-making, and supports organizational growth. The proposed SRS clearly defines the functional and non-functional requirements, actors, use cases, data requirements, assumptions, constraints, and validation criteria, providing a strong foundation for successful system design and implementation.

## Rubrics for Calculation

| Criteria                                                              | Excellent (4 Marks)                                                                                                      | Good (3 Marks)                                                 | Satisfactory (2 Marks)                                 | Needs Improvement (1 Mark)                                     |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------|
| <b>Scenario Understanding and Problem Analysis (30%)</b>              | Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis. | Demonstrates good understanding with minor omissions.          | Basic understanding with limited analysis.             | Poor understanding; major aspects missing or incorrect.        |
| <b>Functional and Non-Functional Requirement Identification (25%)</b> | Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.              | Identifies most requirements with minor classification errors. | Identifies only basic requirements; several omissions. | Requirements are incomplete, inaccurate, or poorly classified. |

|                                                                     |                                                                                                                                                               |                                                                      |                                                                 |                                                                      |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Actor Identification and Use Case Modelling (20%)</b>            | Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.                                    | Most actors and use cases identified; diagram has minor errors.      | Limited actors/use cases identified; diagram partially correct. | Actors/use cases missing or incorrect; diagram absent or inaccurate. |
| <b>Software Requirement Specification (SRS) Documentation (15%)</b> | SRS is well-structured, complete, professionally organized, and includes all required sections.                                                               | SRS is organized with minor missing sections or formatting issues.   | SRS includes only essential sections with limited detail.       | SRS is incomplete, poorly organized, or lacks essential components.  |
| <b>Assumptions, Constraints, and Requirement Validation (10%)</b>   | Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification. | Identifies most assumptions and constraints; validation is adequate. | Limited assumptions/constraints; validation is superficial.     | Assumptions, constraints, and validation are missing or incorrect.   |